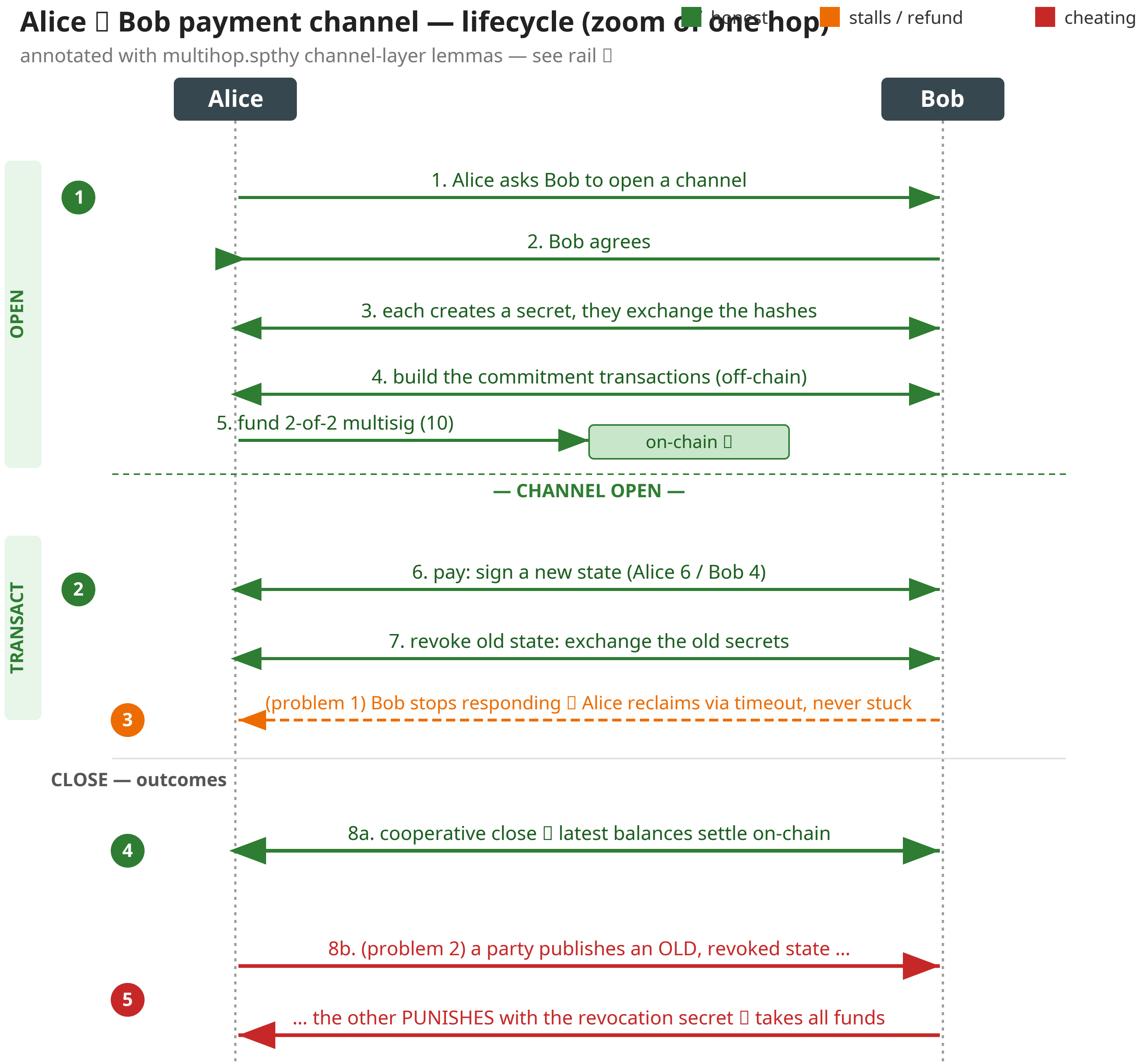
Alice □ Bob payment channel — lifecycle (zoom on one hop)

annotated with multihop.spthy channel-layer lemmas — see rail □



The HTLC attached to this channel (hashlock + timelock) is what the 4-party figure routes; its add consumes Free(ptr) — structurally one HTLC over the modelled channel lifetime.

LEMMAS · channel layer

1 **Channel open & funding**
Protocol_execution □ (witness) · Distinct_Parties_Configuration □
Funds_Locked_Before_Update □ · HTLC_On_Opened_Channel □

2 **State update & revocation**
state_update □ · Update_Requires_Negotiation □
Invoice_Released_Once □ · Ltk_Known_Implies_Compromised □

3 **Refund / anti-stall**
Refund_Requires_Timeout □ · Refund_Possible □ (witness)

4 **Cooperative close & settlement**
Cooperative_Close_Execution □ · settlement_is_traceable □
instant_funds □ · delayed_funds □

5 **Punishment (safety, not an attack)**
No_Punishment_Without_Cheating □
— red trace = the cheat; the lemma proves punishment □ prior cheat

□ = machine-checked in multihop.spthy.
Routing / value / attack lemmas are on the 4-party figure (anchors □–□ there).

Channel-layer lemmas in the fixed-hop abstraction; routing and timing are shown separately.